

Article

Performance Improvement of a Refrigeration Cycle Using the Subcooling Effect from the Thermal Energy Storage

Chittin Tangthieng* and Ratanon Pohnpinyo

Department of Mechanical Engineering, Faculty of Engineering, Chulalongkorn University, Phayathai Rd., Wangmai, Pathumwan, Bangkok 10330, Thailand

E-mail: *QED690@yahoo.com (Corresponding author)

Abstract. This project presents the effect of the subcooling heat rate on the performance of the vapor compression refrigeration cycle. In order to achieve the subcooling effect, the thermal energy storage using an aqueous solution under the sub-zero condition is implemented. A mathematical model of the main refrigeration cycle with the subcooling system is developed. The successive iteration method is applied to obtain the numerical solutions. The input compressor power from the model is compared with the data obtained from the field measurement with a good qualitative agreement. A parametric study of ambient and cold room temperatures at different subcooling heat rates is analyzed. The results reveal that the implementation of the subcooling system is suitable on a main refrigeration cycle working at a lower temperature causing the COP increases at a higher rate. On the other hand, the influence of different ambient temperatures on the performance of the main refrigeration cycle for a given subcooling heat rate is not substantial. If the additional power of the supplemental chiller is included, the results indicate that the subcooling effect is still beneficial at lower cold room temperature. The subcooling system provides no benefit when the cold room temperature is increased to a certain point. Similarly, the benefit from the subcooling system overcomes the additional power from the supplemental chiller leading to the increase of the over coefficient of performance if the ambient temperature is higher to a certain limit.

Keywords: Refrigeration cycle, Subcooling effect, Thermal energy storage, Parametric study

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1. Introduction

Refrigeration cycles are essential in various applications, including air conditioning, product control environment, food preservation, and industrial cooling. Improving their energy efficiency is crucial for reducing operational costs and minimizing environmental impact. One effective method to enhance the performance of a refrigeration cycle is by increasing the degree of subcooling at the condenser outlet. The degree of subcooling is referred to as the temperature difference between the subcooling liquid refrigerant temperature and its saturation temperature at a given pressure after it has fully condensed in the condenser. A higher degree of subcooling ensures that the refrigerant entering the expansion valve is entirely in the liquid state without flash gas formation, leading to the higher refrigeration effect and the improved coefficient of performance.

The effect of the condenser subcooling on the refrigeration cycle has been extensively studied by many researchers. A common method to achieve the subcooling effect is to enhance the heat transfer area of the condenser between the refrigerant and a cooling medium. The related studies showed that the COP after adding the subcooling section to the condenser was increased in a certain level [1,2]. Another method is to implement the liquid suction heat exchanger, which allows the refrigerant liquid to become subcooled by rejecting heat to the refrigerant vapor leaving the evaporator. The comparison of the refrigeration performance between with and without the liquid suction heat exchanger was investigated by both theoretical [3,4] and experimental approaches [5-7]. The results showed that the change of the COP depended on the operating condition and the refrigerant properties. The liquid suction heat exchange can be applied with the mechanical subcooling loop powered by an additional compressor [8,9]. On the other hand, a mechanical subcooling system involving a separated refrigeration cycle from the main one is referred to as the dedicated mechanical subcooling system. The performance of the refrigeration cycles was investigated by numerical modeling [10-12] and experiments [13]. The results indicated that the COP of the refrigeration cycle with a dedicated mechanical subcooling system was still increased despite the addition input power. To reduce this power, the free cooling obtained from either an evaporative cooler or a radiative sky cooling panel was studied [14,15]. The thermal energy storage (TES) can be implemented together with the subcooling system. The experimental investigations of the TES functioned similar to the liquid suction heat exchanger with a delay period were presented [16,17]. The separated TES from the main refrigeration was also studied with the chilled water as a storage medium [18,19]. The limitations of using chilled water are low storage density and not applicable to sub-zero applications. To overcome this limitation, the use of phase change material (PCM) as a storage medium was also investigated in cooperation with the main refrigeration system [20,21]. The drawback when using

the PCM is the complexity of the freezing system and higher initial and operational costs.

Unlike the previous researches in which either chilled water or PCM is used in the TES system, this project aims to study the effect of the subcooling heat rate from a separated thermal energy storage on the performance of the main refrigeration cycle using an aqueous solution. The advantages of the aqueous solution are allowing the TES medium to work under the sub-zero applications and avoiding the complex and costly system. The parametric analysis is conducted to show the appropriate condition for operating the main refrigeration cycle with the TES system. A mathematical model of the main refrigeration cycle, together with the subcooler, is developed. The solution from the model is obtained by employing the successive iteration technique, and compared with the field measurement. A parametric study of different ambient and cold room temperatures is performed. The effects of the heat transfer rate from the subcooler on the main refrigeration cycle are investigated.

2. Description of the Refrigeration System

The system under consideration is an industrial vapor compression system using for a cold storage with an area of 27 x 13 m² of a pilot plant. This cold room has three identical refrigeration units. In each unit, the reciprocating compressor is used to compress the refrigerant of R-22, and then cooled with an air-cooled condenser. The refrigerant is expanded through a thermostatic expansion valve, and fed to a fan coil unit, served as an evaporator, with a cooling capacity of 56 kW. The evaporator supplies cold air to a cold room with the set point temperature of 0–10°C. To implement the subcooling process, the thermal energy storage (TES) system filled with an aqueous solution is installed as shown in Figure 1.

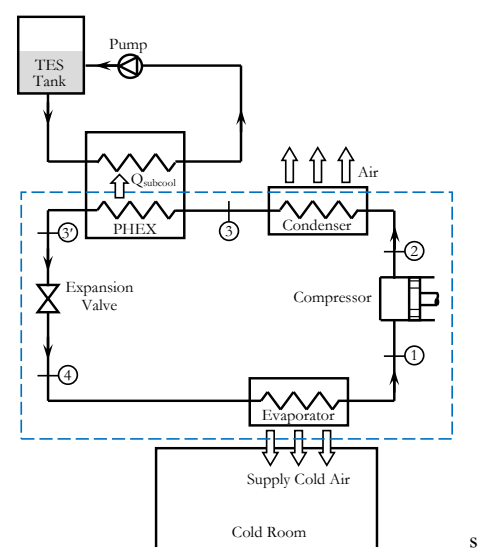


Fig. 1. The schematic diagram of the refrigeration cycle and the subcooler during the on-peak period with the corresponding refrigerant states.

A transfer medium of an aqueous solution is sent from a TES tank to a plate heat exchanger (PHEX), served as a subcooler of the refrigeration cycle. The aqueous solution is chilled and stored in the TES tank during the off-peak period by a supplemental chiller. On the other hand, the aqueous solution is pumped to the PHEX during the on-peak period to provide the subcooling effect to the main refrigeration cycle. The main focus in this study will be on the peak-period where the subcooling effect from the TES tank improves the performance of the main refrigeration cycle.

3. Mathematical Models

Mathematical models of the main refrigeration cycle including its components are developed according to these assumptions. 1) The main refrigeration cycle is under a quasi-steady state with the corresponding T-s diagram of the cycle is depicted in Figure 2. 2) The performance of the compressor at different condensing and evaporating temperatures is provided by the compressor manufacturer. 3) The performances of the condenser and evaporator follow the heat transfer mechanism of the finned-tube cross flow heat exchanger. 4) The thermostatic expansion valve controls the degree of superheat of the refrigerant to be 10 K before entering the compressor. 5) The refrigerant leaving the condenser is in a saturated liquid state. 6) Friction losses in pipes and equipment are neglected. 7) The pipes are well insulated. 8) The thermodynamic properties of R-22 is obtained from [22,23]. From these assumptions the mathematical models for each component can be developed accordingly.

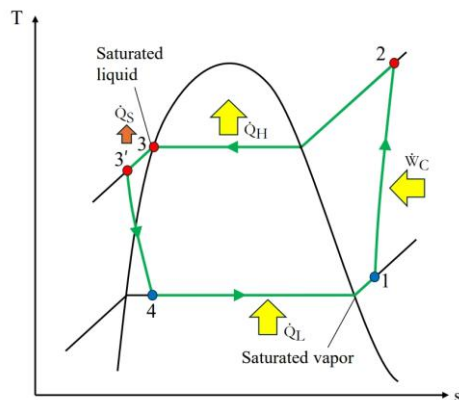


Fig. 2. The T-s diagram of the main refrigeration cycle.

3.1. Compressor

For the reciprocating compressor in this study, its performance curve is obtained by applying the multiple polynomial regression to the manufacturer database presenting in a tabular format. The independent variables are the evaporating temperature (T_o) and condensing temperature (T_c), of which the saturation pressures are corresponding to the suction pressure (P_o) and discharge pressure (P_c), respectively. The dependent variables are the mass flow rate of the refrigerant ($\dot{m}_R$) and the

refrigerant discharge temperature (T_{R2}). The equations for $\dot{m}_R$ and T_{R2} as a function of T_o and T_c are as follows

$$\dot{m}_R = a_0 + a_1 T_o + a_2 T_o^2 + a_3 T_c + a_4 T_o T_c + a_5 T_o^2 T_c \quad (1)$$

$$T_{R2} = a_0 + a_1 T_o + a_2 T_o^2 + a_3 T_c + a_4 T_c^2 + a_5 T_o T_c + a_6 T_o^2 T_c + a_7 T_o T_c^2 + a_8 T_o^2 T_c^2 \quad (2)$$

The polynomial coefficients (a_0 to a_8) are given in Table 1. The input power of the compressor ($\dot{W}_C$) can be written from the energy balance at the refrigeration cycle.

$$\dot{W}_C = \dot{m}_R (h_{R2} - h_{R1}) \quad (3)$$

Table 1. The polynomial coefficient of refrigerant mass flow rate and the discharge temperature

Coefficients	Values for $\dot{m}_R$	Values for T_{R2}
a_0	4.0317E-01	2.8216E+01
a_1	1.3887E-02	-6.0169E-01
a_2	1.4366E-04	-1.6105E-03
a_3	-1.3993E-03	1.3796
a_4	-3.6651E-05	5.2031E-04
a_5	-4.2966E-07	5.2335E-03
a_6	-	8.8714E-04
a_7	-	-2.3962E-04
a_8	-	-7.7946E-06

3.2. Condenser and Evaporator

The heat transfer rate of the condenser and evaporator can be described by the thermal performance of the finned tube heat exchanger. The effectiveness NTU method is used for the heat transfer analysis.

$$\dot{Q} = \epsilon \times \dot{Q}_{\max} = C_{\min} \times (T_{h,in} - T_{c,in}) \quad (4)$$

where $\dot{Q}$ is actual heat transfer rate, $\dot{Q}_{\max}$ is maximum heat transfer rate, ϵ is effectiveness, $C_{\min}$ is smaller heat capacity rate between hot and cold fluids, $T_{h,in}$ is hot fluid inlet temperature and $T_{c,in}$ is cold fluid inlet temperature. In case of the condenser, $T_{h,in}$ and $T_{c,in}$ is the condensing temperature (T_c) and the ambient temperature (T_{amb}), respectively. On the other hand, for the evaporator, $T_{h,in}$ and $T_{c,in}$ is the cold room temperature (T_{room}) and the evaporating temperature (T_o), respectively. The relations between the effectiveness and the number of transfer unit of NTU for the condenser and evaporator involving the phase-change process are given by [24].

$$NTU = \frac{UA}{C_{\min}} \quad (5)$$

$$\epsilon = 1 - \exp(-NTU) \quad (6)$$

The overall heat transfer coefficient (UA) is calculated from the combination of the heat transfer coefficients of the refrigerant inside the tubes ($h_{R,i}$) and air flowing

through the finned coil outside ($h_{a,o}$) with the corresponding areas A_i and A_o , respectively. Note that the thermal resistance of the heat conduction through the tube wall is neglected.

$$\frac{1}{(UA)} = \frac{1}{h_{R,i} \cdot A_i} + \frac{1}{h_{a,o} \cdot A_o} \quad (7)$$

The heat rejection rate of the refrigeration cycle ($\dot{Q}_H$) and the cooling load of the refrigeration cycle ($\dot{Q}_L$) are also satisfied by the energy balances at the condenser and evaporator, respectively.

$$\dot{Q}_H = \dot{m}_R (h_{R2} - h_{R3}) \quad (8)$$

$$\dot{Q}_L = \dot{m}_R (h_{R1} - h_{R4}) \quad (9)$$

The coefficient of performance of the main refrigeration cycle (COP_R) is calculated from the following equation.

$$COP_R = \frac{\dot{Q}_L}{\dot{W}_C} \quad (10)$$

Tables 2 and 3 provide the specifications of the condenser and evaporator, respectively.

Table 2. Condenser specifications

Description	Value	Unit
Number of fans	2	unit
Face area	0.6616	m ²
Face air velocity	6.16	m/s
Number of tubes	184	tubes
Tube outside diameter	0.00953	m
Tube inside diameter	0.00775	m
Tube thickness	0.00076	m
Tube length	2.3	m
Heat transfer coefficient outside the tubes	598	W/m ² -K
Heat transfer coefficient inside the tubes	7565	W/m ² -K

Table 3. Evaporator specifications

Description	Value	Unit
Number of fans	3	unit
Face area	0.2590	m ²
Face air velocity	6	m/s
Number of tubes	256	tubes
Tube outside diameter	0.00953	m
Tube inside diameter	0.00775	m
Tube thickness	0.00076	m
Tube length	2.99	m
Heat transfer coefficient outside the tubes	245	W/m ² -K
Heat transfer coefficient inside the tubes	4040	W/m ² -K

These include geometries of both and air velocities. The heat transfer coefficients for the R-22 flow inside the tube for the condenser and evaporator are estimated by Ref. [25] and [26], respectively. On the other hand, the heat transfer coefficients for the air flow through the fins outside the tube for both condenser and evaporator are estimated based on Gray and Webb correlation [27].

3.3. Plate Heat Exchanger

The plate heat exchanger (PHEX) is served as a subcooler causing the liquid refrigerant temperature dropping below the saturation temperature from state 3 to the subcooling state 3' as shown in Figure 2. The subcooling effect stored within the TES tank is generated by the supplemental chiller during the off-peak period as shown in Figure 3(a). During the off-peak period, no subcooling effect is provided to the refrigeration cycle. On the other hand, during the on-peak period, the subcooling effect will be utilized by circulating the aqueous solution from the TES tank to the PHEX by a pump depicted in Figure 3(b). The amount of subcooling heat rate ($\dot{Q}_{Sc}$) can be determined by considering a system surrounding by a dotted line as shown in Figure 3(b).

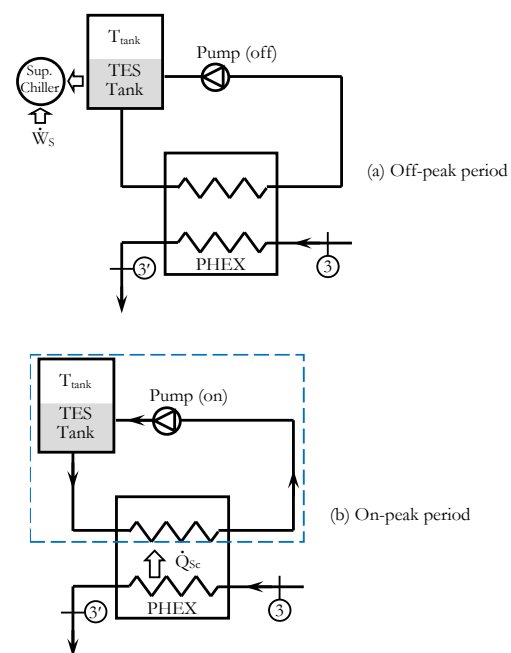


Fig. 3. The schematic diagram of the TES tank and the subcooler during the (a) off-peak period and (b) on-peak period with the corresponding refrigerant states.

During the on-peak period, the temperature of the aqueous solution inside the tank (T_{tank}) will gradually increase from approximately -3 to 5.6°C due to $\dot{Q}_{Sc}$ at the PHEX. By assuming the steady-state mass flow rate through the pump, the negligible amount of pump power, and the negligible heat loss through the pipe, $\dot{Q}_{Sc}$ can be determined by calculating the rate of increasing temperature in the TES tank with respect to time as written in Eq. (11). Although the mass in the TES tank is

under the steady-state condition, the energy in the TES tank is not.

$$\dot{Q}_{Sc} = m_{sol} C_{sol} \frac{dT_{tank}}{dt} \quad (11)$$

In Eq. (11), m_{sol} is mass of the aqueous solution in the TES tank of 9,600 kg. C_{sol} is the specific heat of the aqueous solution of 3.74 kJ/kg-K. When $\dot{Q}_{Sc}$ has been determined, state 3' is prescribed by the energy balance at the subcooler.

$$\dot{Q}_{Sc} = \dot{m}_R(h_{R3} - h_{R3'}) \quad (12)$$

It is noted that in case of the absence of $\dot{Q}_{Sc}$, state 3' is identical to state 3. To complete the refrigeration cycle, the enthalpy of state 4 is equal to the enthalpy of state 3' due to the expansion through the expansion valve.

From Figure 3(a), it can be seen that the subcooling energy stored in the TES tank requires an additional power ($\dot{W}_S$) from the supplemental chiller. It will be added in order to calculate the overall coefficient of performance (COP_T).

$$COP_T = \frac{\dot{Q}_L}{\dot{W}_C + \dot{W}_S} \quad (13)$$

Since the specification data of the supplemental chiller is very limited, a constant coefficient of performance of the supplemental chiller (COP_S) is assumed to be 2.684. The value of this COP_S is adjusted by using the correction factor calculated based on T_{room} and T_{amb} . As a result, the COP_T can be written as

$$COP_T = \frac{\dot{Q}_L}{\dot{W}_C + (\dot{Q}_{Sc}/COP_S)} \quad (14)$$

The COP_T represents the overall performance of the refrigeration cycle with the subcooling effect. The investigation of the effect of $\dot{Q}_{Sc}$ on the COP_T will be provided.

4. Solution Procedure and Verification

The solution procedure for the solution requires the iterative method. Figure 4 illustrates the flowchart diagram of the solution procedure. The multivariate quadratic regression method is used to determine the enthalpy of R-22 as a function of given thermodynamic properties, such as P and/or T, for saturated, superheat and subcooled regions. The bisection method is applied due to its guaranteed convergence. The iterative processes continue until the convergence criteria for error tolerance of 10^{-6} are satisfied. In this study, the variation of the performance of the refrigeration cycle with different subcooling heat rates ($\dot{Q}_{Sc}$) is investigated at different T_{room} and T_{amb} . The performance of the cycle includes $\dot{m}_R$, $\dot{W}_C$, $\dot{Q}_L$, and the COP.

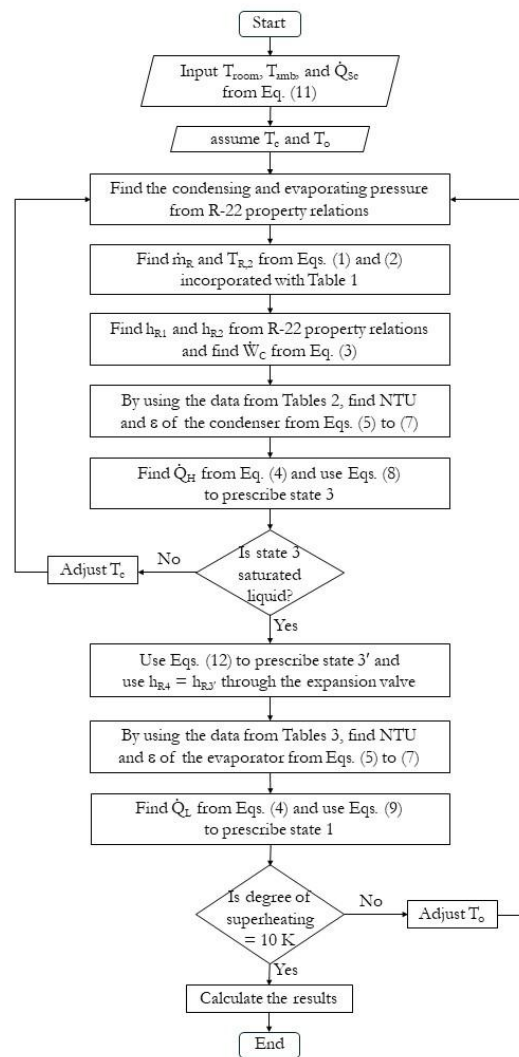


Fig. 4. The flowchart diagram of the solution procedure.

The prediction from the model is verified by the field measurement at the pilot plant. As a normal operation, the set point temperature for the cold room is 0°C. This temperature could be fluctuating up depending on the loading condition. On the measuring period of 24 hour, the ambient temperature is between 30.0-34.5°C. To measure T_{tank} , T_{room} , and T_{amb} , the type-K thermocouple sensors are placed in three different locations, and the temperature data is obtained in every minutes. The value of T_{tank} is used to calculate $\dot{Q}_{Sc}$ based on Eq. (11) where dT_{tank}/dt is estimated by using the central difference approximation. The hourly average is performed to reduce the effect of short-term variations. The measuring results are shown in Table 4.

Table 4. Data from the field measurement

Case	$\dot{Q}_{Sc}$ (kW)	T_{room} (°C)	T_{amb} (°C)
1	1.99	2.0	33.9
2	3.57	1.9	33.8
3	5.19	1.4	31.5
4	6.82	1.5	33.5
5	8.47	1.7	34.1
6	9.85	2.6	33.4
7	11.32	2.4	33.7
8	12.80	1.6	31.1

The input power of the compressor ($\dot{W}_C$) can be directly measured by the power meter with data logger providing ± 1 % accuracy. The values of $\dot{W}_C$ are used for validation of the model prediction.

5. Results and Discussion

The results obtained from the refrigeration model are verified by the field measurement by comparing the values of $\dot{W}_C$ as depicted in Figure 5. It is noted that $\dot{W}_C$ obtained from the model is calculated based on the measurement values of T_{room} , and T_{amb} tabulated in Table 4. The qualitative agreement between the values of $\dot{W}_C$ obtained from both methods is observed. It can be seen that $\dot{W}_C$ of 12.8 to 14.0 kW obtained from the model prediction is higher than $\dot{W}_C$ of 10.1 to 11.8 kW obtained from the field measurement. The relative error between those for all eight cases are between 18.3% and 27.0%. The overprediction by the model stems from the degradation of the actual refrigeration parts that have been operated for more than 20 years leading to the lower performance.

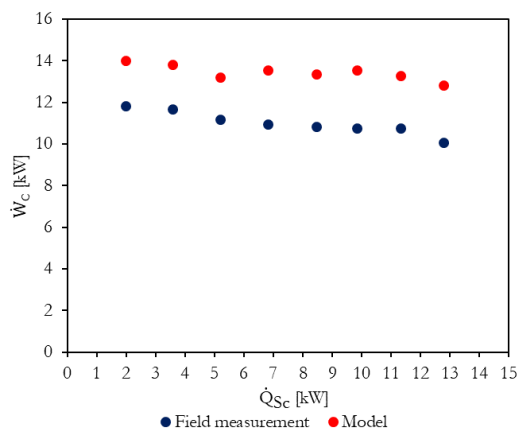


Fig. 5. Comparison of the input power between the numerical prediction and field measurement

The parametric study of the refrigeration model is performed by investigation of the effect of $\dot{Q}_{Sc}$ from 0 to 15 kW on $\dot{m}_R$, $\dot{W}_C$, $\dot{Q}_L$, and the COP at different T_{room} and T_{amb} . The ranges of T_{room} and T_{amb} are from -10 to 10°C and 25 to 40°C , respectively. Figure 6 represents the variation of $\dot{m}_R$ with respect to $\dot{Q}_{Sc}$ at different T_{room} at

constant T_{amb} of 35°C . At a given T_{room} , an increase of $\dot{Q}_{Sc}$ will raise the total cooling capacity of the refrigeration cycle causing T_o and P_o to drop. As a result, the compressor pressure ratio (P_c/P_o) will increase, leading to a lower value of $\dot{m}_R$. As $\dot{Q}_{Sc}$ is increased from 0 to 15 kW, $\dot{m}_R$ decreases by approximately 14.0%, 12.2%, and 10.5% at T_{room} of -10 , 0, and 10°C , respectively. This indicates that the variation of $\dot{m}_R$ with the change of $\dot{Q}_{Sc}$ is less sensitive in case of higher T_{room} .

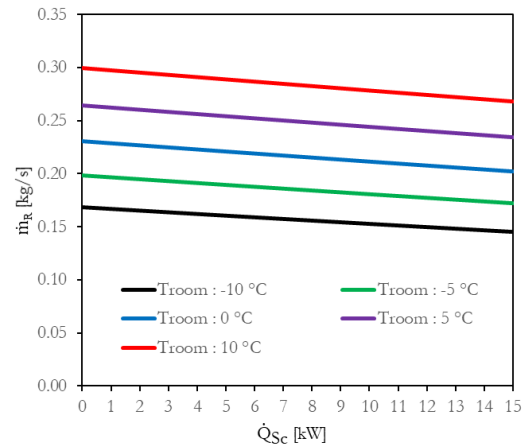


Fig. 6. Variation of the refrigerant mass flow rate with different room temperatures at the ambient temperature of 35°C .

The variation of $\dot{m}_R$ with respect to $\dot{Q}_{Sc}$ at different T_{amb} at constant T_{room} of 0°C is illustrated in Figure 7. As a given T_{amb} , an increase of $\dot{Q}_{Sc}$ results in the higher compressor pressure ratio and the lower $\dot{m}_R$ similar to what has been observed in Figure 6. As $\dot{Q}_{Sc}$ is increased from 0 to 15 kW, $\dot{m}_R$ decreases by approximately 11.8%, 11.9%, 12.1% and 12.3% at T_{amb} of 25, 30, 35, and 40°C , respectively. This indicates that the variation of $\dot{m}_R$ with the change of $\dot{Q}_{Sc}$ is not sensitive to the change of T_{amb} .

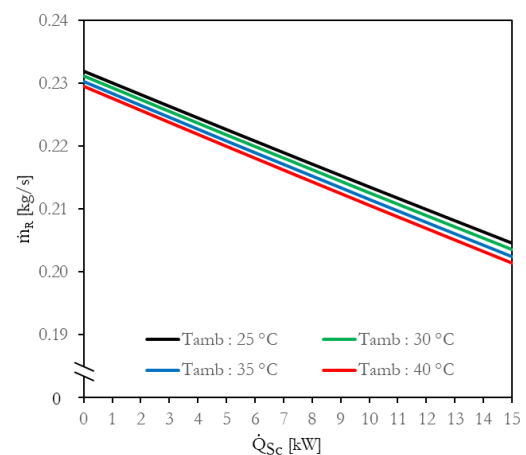


Fig. 7. Variation of the refrigerant mass flow rate with different ambient temperatures at the room temperature of 0°C .

Figure 8 depicts the variation of $\dot{Q}_L$ with respect $\dot{Q}_{Sc}$ at different T_{room} at constant T_{amb} of 35°C. For a given T_{room} , an increase of $\dot{Q}_{Sc}$ leads to the higher degree of subcooling, reducing the value of h_{R3} further. As a result, the refrigeration effect, i.e., $h_{R1} - h_{R4}$, is significantly higher, which overcomes the effect of the reduction of $\dot{m}_R$. Consequently, the value of $\dot{Q}_L$ from Eq. (9) is higher. As $\dot{Q}_{Sc}$ is increased from 0 to 15 kW, $\dot{Q}_L$ increases by approximately 44.4%, 30.8%, and 22.7% at T_{room} of -10, 0, and 10°C, respectively. The variation of $\dot{Q}_L$ with the change of $\dot{Q}_{Sc}$ is highly sensitive at a lower value of T_{room} . This result leads to a suggestion to apply a subcooler to a lower-temperature refrigeration application, rather than an A/C one.

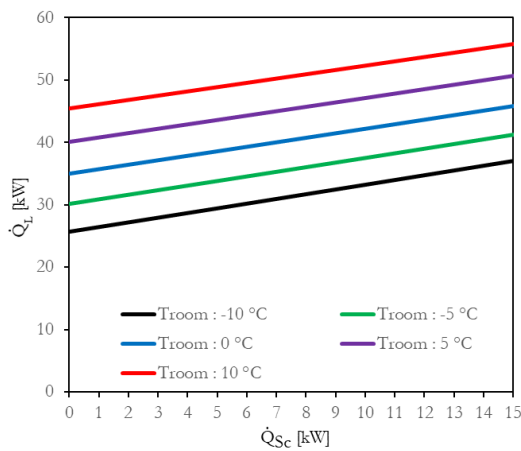


Fig. 8. Variation of the cooling capacity with different room temperatures at the ambient temperature of 35°C.

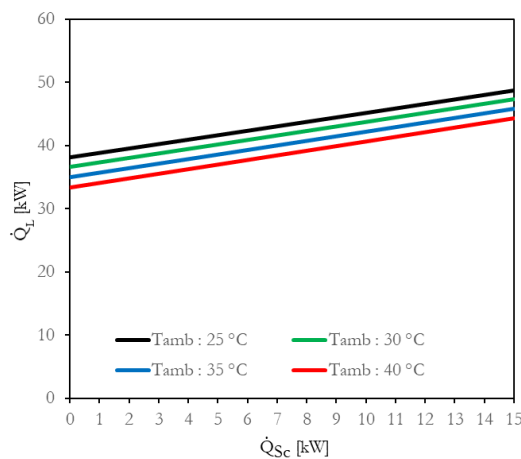


Fig. 9. Variation of the cooling capacity with different ambient temperatures at the room temperature of 0°C.

The variation of $\dot{Q}_L$ with respect to $\dot{Q}_{Sc}$ at different T_{amb} at constant T_{room} of 0°C is illustrated in Figure 9. For a given T_{amb} , an increase of $\dot{Q}_{Sc}$ leads to the higher $\dot{Q}_L$ similar to what is observed in Figure 8. As $\dot{Q}_{Sc}$ is increased from 0 to 15 kW, $\dot{Q}_L$ increases by approximately 27.6%, 29.1%, 30.8%, and 32.7% at T_{amb} of 25, 30, 35, and 40°C, respectively. The variation of $\dot{Q}_L$ with the change

of $\dot{Q}_{Sc}$ is slightly more sensitive in case of higher T_{amb} . The application of the subcooler to gain more cooling capacity will be slightly more effective at a region or a season with a higher T_{amb} .

Figure 10 illustrates the variation of $\dot{W}_C$ with respect $\dot{Q}_{Sc}$ at different T_{room} at constant T_{amb} of 35°C. For a given T_{room} , an increase of $\dot{Q}_{Sc}$ leads to lower T_o , resulting in a higher value of $h_{R2} - h_{R1}$. However the reduction of $\dot{m}_R$ as shown in Figure 6 dominates an increase of $h_{R2} - h_{R1}$, $\dot{W}_C$ will slightly decrease based on Eq. (3) accordingly. As $\dot{Q}_{Sc}$ is increased from 0 to 15 kW, $\dot{W}_C$ decreases by approximately 8.16%, 6.47%, and 5.29% at T_{room} of -10, 0, and 10°C, respectively. The variation of $\dot{W}_C$ with the change of $\dot{Q}_{Sc}$ is moderately more sensitive in case of a lower value of T_{room} . Hence, to implement the subcooling effect, a lower temperature application is preferable to a higher temperature one in order to save the more electrical power.

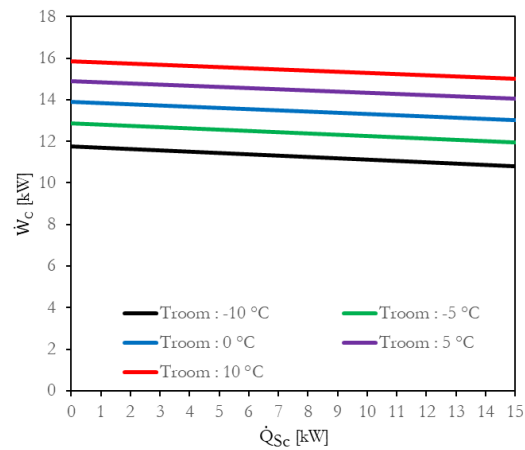


Fig. 10. Variation of the compressor power with different room temperatures at the ambient temperature of 35°C.

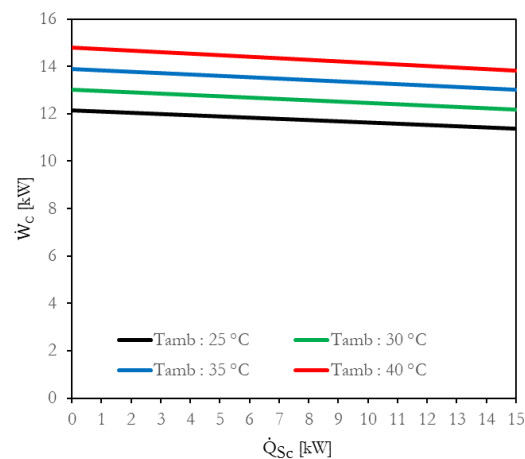


Fig. 11. Variation of the compressor power with different ambient temperatures at the room temperature of 0°C.

The variation of $\dot{W}_C$ with respect to $\dot{Q}_{Sc}$ at different T_{amb} at constant T_{room} of 0°C is illustrated in Figure 11. For a given T_{amb} , an increase of $\dot{Q}_{Sc}$ leads to the lower $\dot{W}_C$

similar to what is observed in Figure 10. As $\dot{Q}_{Sc}$ is increased from 0 to 15 kW, $\dot{W}_C$ decreases by approximately 6.30%, 6.38%, 6.47%, and 6.57% at T_{amb} of 25, 30, 40, and 40°C, respectively. It is observed that the variation of $\dot{W}_C$ with the change of $\dot{Q}_{Sc}$ is marginally sensitive to different values of T_{amb} .

Figure 12 depicts the variation of the COP_R with respect $\dot{Q}_{Sc}$ at different T_{room} at constant T_{amb} of 35°C. For a given T_{room} , according to Figures 8 and 10 by comparing the different values of $\dot{Q}_{Sc}$, the higher subcooling effect increases $\dot{Q}_L$ while decreasing $\dot{W}_C$, resulting in a significant increase of the COP_R . As $\dot{Q}_{Sc}$ is increased from 0 to 15 kW, the COP_R increases by approximately 57.3%, 39.9%, and 29.6% at T_{room} of -10, 0, and 10°C, respectively. However, the variation of the COP_R with the change of $\dot{Q}_{Sc}$ is more sensitive in case of lower T_{room} . As a result, when the subcooling system is applied to the refrigeration cycle, the improvement of the original refrigeration cycle performance with low T_{room} is preferable to that with high T_{room} .

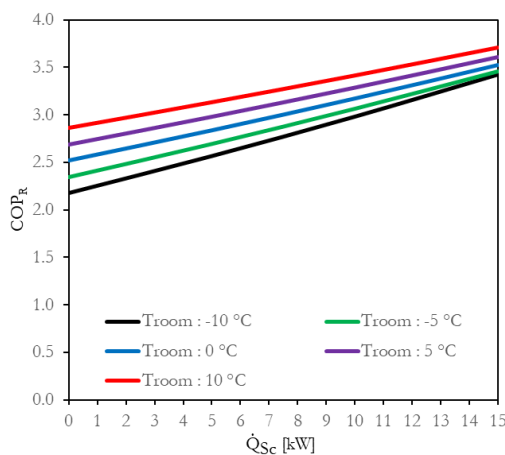


Fig. 12. Variation of the COP_R with different room temperatures at the ambient temperature of 35°C.

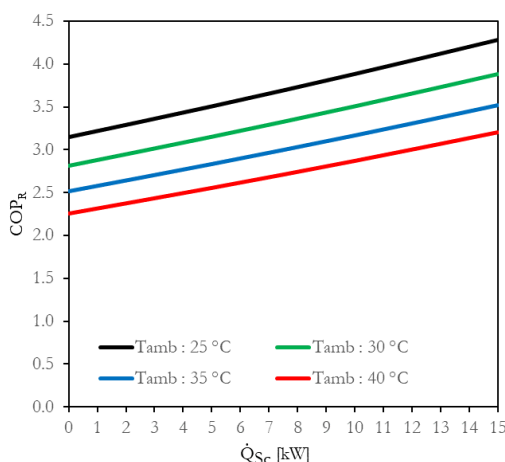


Fig. 13. Variation of the COP_R with different ambient temperatures at the room temperature of 0°C.

The variation of the COP_R with respect to $\dot{Q}_{Sc}$ at different T_{amb} at constant T_{room} of 0°C is presented in Figure 13. According to Figures 9 and 11, for a given of T_{amb} , the higher $\dot{Q}_{Sc}$, the higher COP_R similar to what has been observed in the previous figure. As $\dot{Q}_{Sc}$ is increased from 0 to 15 kW, COP_R is increased by approximately 36.1%, 37.9%, 39.9%, and 42.1% at T_{amb} of 25, 30, 35, and 40°C, respectively. This results indicates that the the variation of the COP_R with the change of $\dot{Q}_{Sc}$ is marginally sensitive to the change of T_{amb} . The subcooling system has a slightly higher improvement rate to the original refrigeration cycle performance in case of a higher T_{amb} .

Figure 14 presents the variation of the COP_T with respect $\dot{Q}_{Sc}$ at different T_{room} at constant T_{amb} of 35°C. By combining the additional power from the supplemental chiller, it can be seen that the trend of COP_T is different depending on the value of T_{room} . At T_{room} of -10 and -5°C, the COP_T increase with increasing $\dot{Q}_{Sc}$ from 0 to 15 kW by 3.69% and 0.38%, respectively. On the other hand, at T_{room} of 0, 5, and 10°C, the COP_T decrease with increasing $\dot{Q}_{Sc}$ by 2.17%, 4.11%, and 5.57%, respectively. This result indicates that the additional power from the supplement chiller will offset the benefit of the higher COP_R shown in Figure 12 at higher T_{room} , leading to the reduction of the COP_T . However, because the benefit from the subcooling effect increases at lower T_{room} , the COP_T is still slightly increases. It is noted that this result highly depends on the COP_s of the supplemental chiller. If the value of COP_s is much lower than the assumed value of 2.846, the additional power from the supplemental chiller could dominate the overall performance with no benefit from the subcooling system.

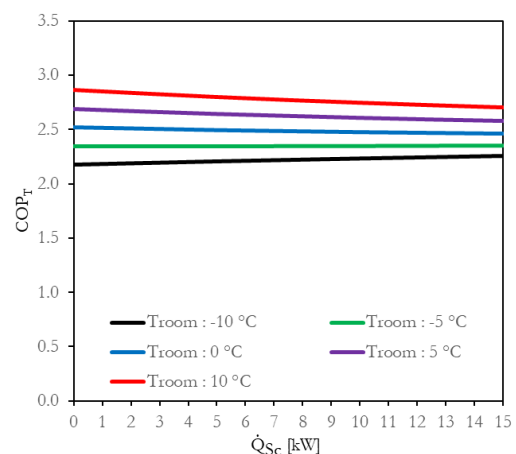


Fig. 14. Variation of the COP_T with different room temperatures at the ambient temperature of 35°C.

The variation of the COP_T with respect to $\dot{Q}_{Sc}$ at different T_{amb} at constant T_{room} of 0°C is presented in Figure 15. The value of the COP_T is decreased by 8.70%, 5.44%, and 2.17% with increasing $\dot{Q}_{Sc}$ from 0 to 15 kW at T_{amb} of 25, 30, and 35°C, respectively. This result indicates that the additional power from the supplemental chiller

overcomes the beneficial effect of the subcooling system. However, the COP_T starts increasing by 1.17% with increasing $\dot{Q}_{Sc}$ from 0 to 15 kW at T_{amb} of 40°C. At high T_{amb} , the subcooling system starts providing the benefit to the overall performance of the system.

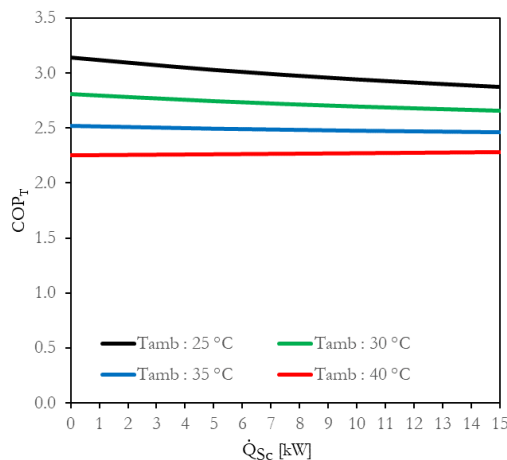


Fig. 15. Variation of the COP_T with different ambient temperatures at the room temperature of 0°C.

6. Conclusions

The study provides the parametric analysis of the subcooling effect on the performance of the main refrigeration cycle at different cold room and ambient temperatures. A mathematical model of the main refrigeration cycle, together with the subcooler, is developed and presented. The successive iteration technique is used to search for the solutions. The data from the field measurement is compared with the solutions obtained from the model prediction with a qualitative agreement. A parametric analysis of the subcooling effect at the cold room temperatures from -10 to 10°C and the ambient temperatures from 25 to 40°C is performed. By considering the main refrigeration cycle alone, as the subcooling heat rate is increases from 0 to 15 kW, the COP_R is increased by 57.3% and 29.6% at the cold room temperature of -10 and 10°C, respectively. On the other hand, as the subcooling heat rate is increased from 0 to 15 kW, the COP_R is increased by 36.1% and 42.1% at the ambient temperature of 25 and 40°C, respectively. The results indicate that the effect of the subcooling system on the performance of the main refrigeration cycle is higher in case of low cold room temperature. Thus, the implementation of the subcooling system will have a higher impact on lower temperature applications. Conversely, the effect of the subcooling system on the performance of the main refrigeration cycle is less substantial in case of the different ambient temperatures. When the additional power from the supplemental chiller is included, as the subcooling heat rate is increases from 0 to 15 kW, the COP_T is increased by 3.69% and decreased by 5.57% at the cold room temperature of -10 and 10°C, respectively. On the other hand, as the subcooling heat rate is increased from 0 to 15 kW, the COP_T is decreased

by 8.70% and increased by 1.17% at the ambient temperature of 25 and 40°C, respectively. This results lead to the conclusion that the subcooling effect is still beneficial at lower cold room temperature. When the cold room temperature is increased to a certain point, no benefit is achieved from the subcooling system. This similar result is observed for the ambient temperature. When the ambient temperature is high enough, the benefit from the subcooling system overcomes the additional power from the supplemental chiller leading to the increase of the over coefficient of performance.

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Chittin Tangthieng was born in Bangkok, Thailand. He received the B.Eng. degree in mechanical engineering from Chulalongkorn University, Thailand, in 1994. He completed his master degree in mechanical engineering from University of Washington, Seattle, WA, USA, in 1998, and earned his Ph.D. in mechanical engineering from the Pennsylvania State University, State College, PA, USA, in 2002. He is currently an associate professor in the department of mechanical engineering, Chulalongkorn University. His research interests include energy management in buildings and industries, refrigeration and air conditioning, and modeling and simulation of thermal power plants



Ratanon Pohnpinyo was born in Samut Prakan, Thailand. He received the bachelor's degree in mechanical engineering from Kasetsart University, Kamphaeng Saen Campus, Thailand, in 2019. He worked as a Production Engineer in an automotive-part manufacturing industry from 2019 to 2022. He is currently pursuing the master's degree in mechanical engineering at Chulalongkorn University, Thailand.